# ARTIN APPROXIMATION REALIZES A VERSAL RING AS A COMPLETED LOCAL RING OF A SMOOTH MORPHISM

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*Editorial note.* All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-geometry.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

*Editorial note.* Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for stacks-geometry.tex (ge-occupied.json, 2026-10-14); 207 route=geometry / chapter-list field hits in focused-seeds are alias noise (route name and alphabetical chapter enumerations), excluded per the ladder alias-blind-spot caveat.

*Source reference.* Original reference 0DQU: sources/stacks-chow/tags.txt, source lines 17199–17199. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 07X8: sources/stacks-chow/tags.txt, source lines 10068–10068. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 0DQJ: sources/stacks-chow/tags.txt, source lines 17189–17189. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 0CMY: sources/stacks-chow/tags.txt, source lines 15873–15873. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 07XH: sources/stacks-chow/tags.txt, source lines 10077–10077. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original lemma 0DQZ: sources/stacks-chow/tags.txt, source lines 17204–17204. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

**Lemma 0.1.** *In Situation 0DQU. Let $x_0 : \mathrm{Spec}(k) \to \mathcal{X}$ be a morphism such that $\mathrm{Spec}(k) \to S$ is of finite type with image $s$. Let $A$ be a versal ring to $\mathcal{X}$ at $x_0$. If $\mathcal{O}_{S,s}$ is a G-ring, then we may find a smooth morphism $U \to \mathcal{X}$ whose source is a scheme and a point $u_0 \in U$ with residue field $k$, such that*

(1) *$\mathrm{Spec}(k) \to U \to \mathcal{X}$ coincides with the given morphism $x_0$,*
(2) *there is an isomorphism $\mathcal{O}^\wedge_{U,u_0} \cong A$.*

*Date*: stacks-project:stacks-geometry.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Proof.* Let $(\xi_n, f_n)$ be the versal formal object over $A$. By Artin's Axioms, Lemma 07X8 we know that $\xi = (A, \xi_n, f_n)$ is effective. By assumption $\mathcal{X}$ is locally of finite presentation over $S$ (use Morphisms of Stacks, Lemma 0DQJ), and hence limit preserving by Limits of Stacks, Proposition 0CMY. Thus Artin approximation as in Artin's Axioms, Lemma 07XH shows that we may find a morphism $U \to \mathcal{X}$ with source a finite type $S$-scheme, containing a point $u_0 \in U$ of residue field $k$ satisfying (1) and (2) such that $U \to \mathcal{X}$ is versal at $u_0$. By Lemma 0DQZ after shrinking $U$ we may assume $U \to \mathcal{X}$ is smooth. $\square$